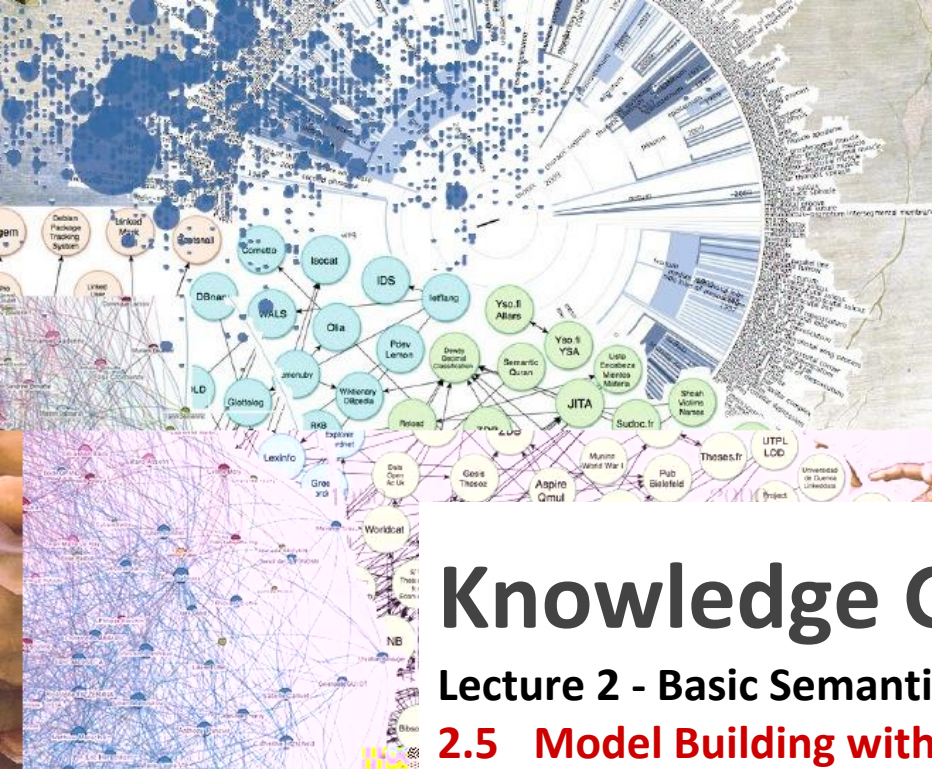




Knowledge Graphs

Lecture 2 - Basic Semantic Technologies

2.5 Model Building with RDFS

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Knowledge Graphs

Lecture 2: Basic Semantic Technologies

2.1 How to Identify and Access Things

2.2 How to Represent Simple Facts with RDF

2.3 RDF Turtle Serialization

2.4 RDF Complex Data Structures

2.5 Model Building with RDFS

2.6 Logical Inference with RDF(S)

Excursion 1: RDFa - RDF and the Web

What does it really mean?

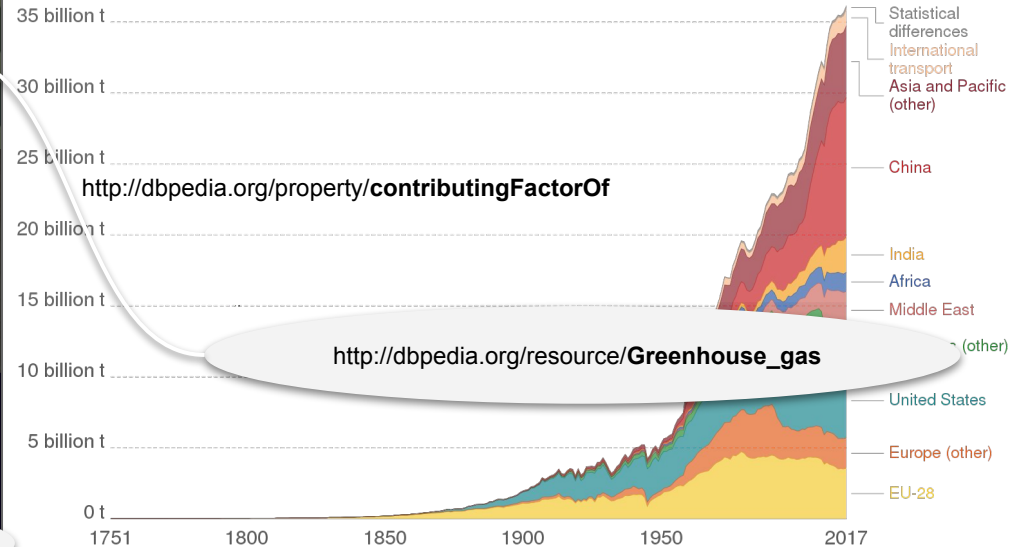
http://dbpedia.org/resource/Greenhouse_effect

<http://dbpedia.org/property/contributingFactorOf>

<http://dbpedia.org/ontology/discoverer>

http://dbpedia.org/resource/Joseph_Fourier

Annual total CO₂ emissions, by world region

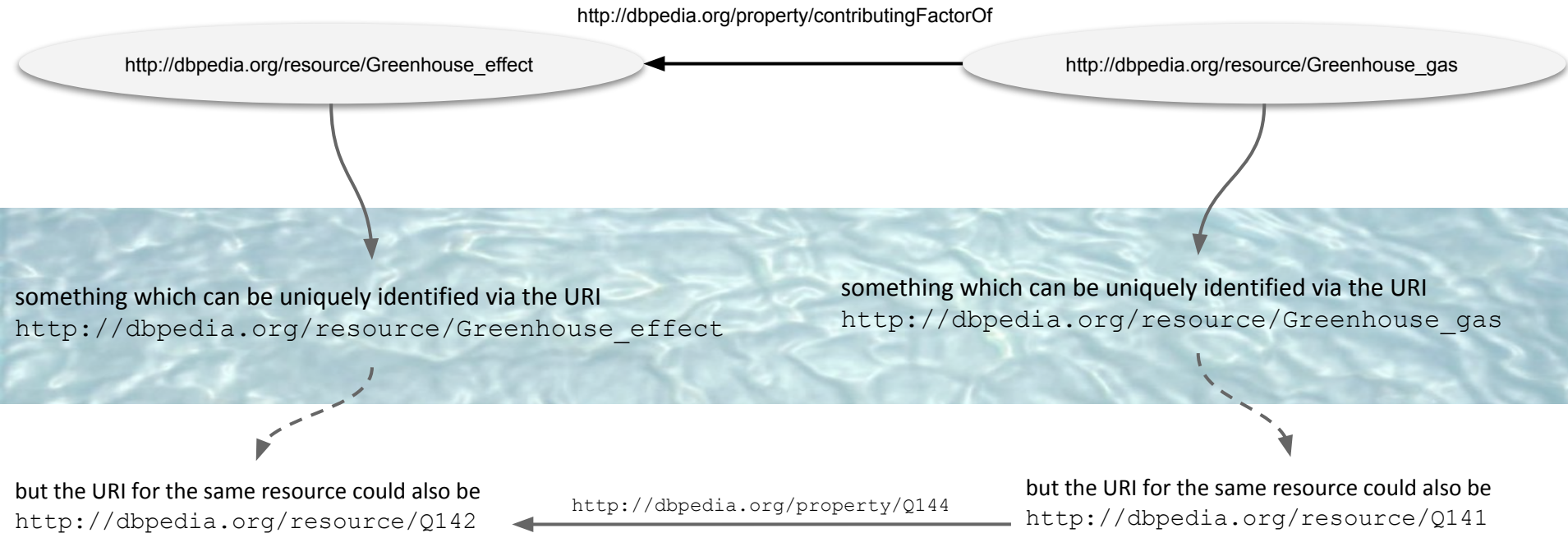


Source: Carbon Dioxide Information Analysis Center (CDIAC); Global Carbon Project (GCP)

Note: "Statistical differences" notes the discrepancy between estimated global emissions and the sum of all national and international transport emissions.

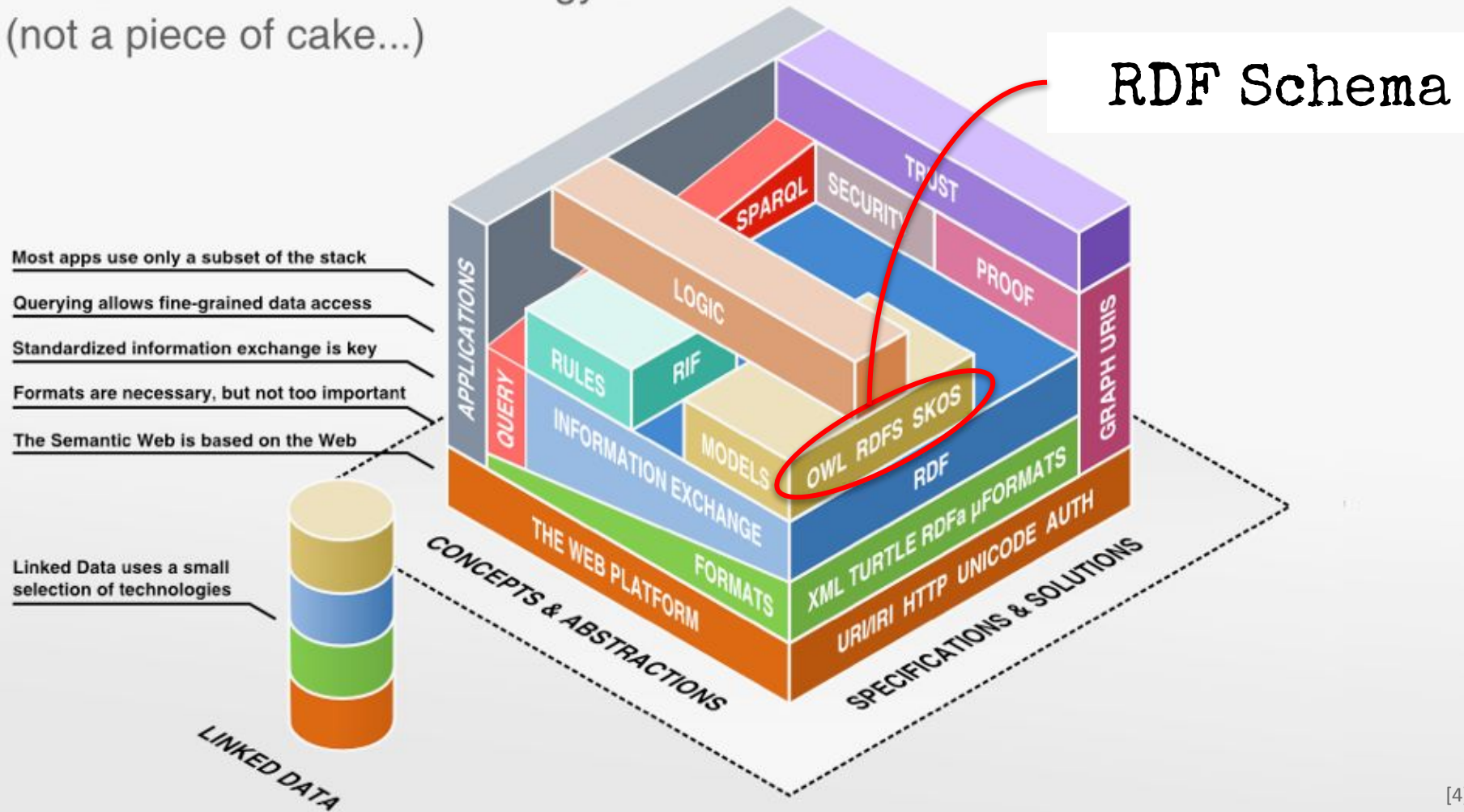
Where does the intended meaning really come from?

What does it really mean?



We need more semantic expressivity...

The Semantic Web Technology Stack (not a piece of cake...)





RDF Schema

- **RDF Schema**, officially called “**RDF Vocabulary Description Language**”
- RDF Schema allows:
 - Definition of **classes** via **rdfs:Class**
 - Class instantiation in RDF via **rdf:type**
 - Example:

```
:Greenhouse_gas    rdf:type rdfs:Class .  
:Carbon_dioxide    rdf:type :Greenhouse_gas .
```

Carbon_dioxide ∈ Greenhouse_gas

```
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .  
@prefix rdf:  <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .  
@prefix :    <http://example.org/Space#> .
```



RDF Schema

- Definition of **properties** via `rdf:Property`
- Definition of **property restrictions on domain and range** via `rdfs:domain` and `rdfs:range`
- Example

```

:Person      rdf:type      rdfs:Class .
:Thing       rdf:type      rdfs:Class .
:discoverer  rdf:type      rdf:Property .
:discoverer  rdfs:domain   :Thing .
:discoverer  rdfs:range    :Person .
  
```

$\text{discoverer} \subseteq \text{Thing} \times \text{Person}$



RDF Schema

- Everything in the RDF model is a **resource**
 - `rdfs:Class` `rdf:type` `rdfs:Resource` .
 - `rdf:Property` `rdf:type` `rdfs:Resource` .
 - `rdfs:Literal` `rdf:type` `rdfs:Resource` .
 - `rdfs:XMLLiteral` `rdf:type` `rdfs:Resource` .
 - `rdfs:Datatype` `rdf:type` `rdfs:Resource` .

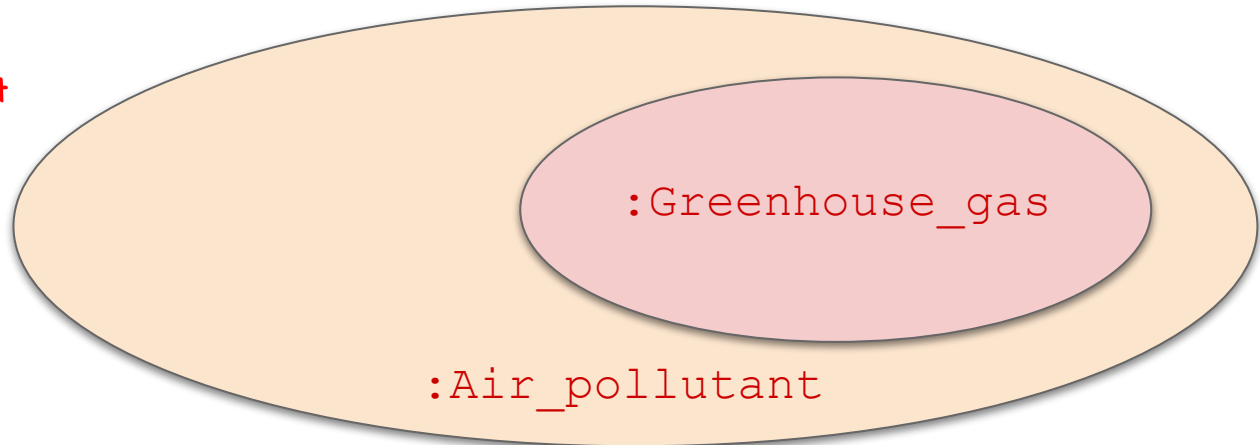


RDF Schema

- Definition of **hierarchical relationships**:
 - **Subclasses** and **superclasses** via `rdfs:subClassOf`
 - Example:

`:Greenhouse_gas rdfs:subClassOf :Air_pollutant .`

$\text{greenhouse_gas} \subseteq \text{Air_pollutant}$



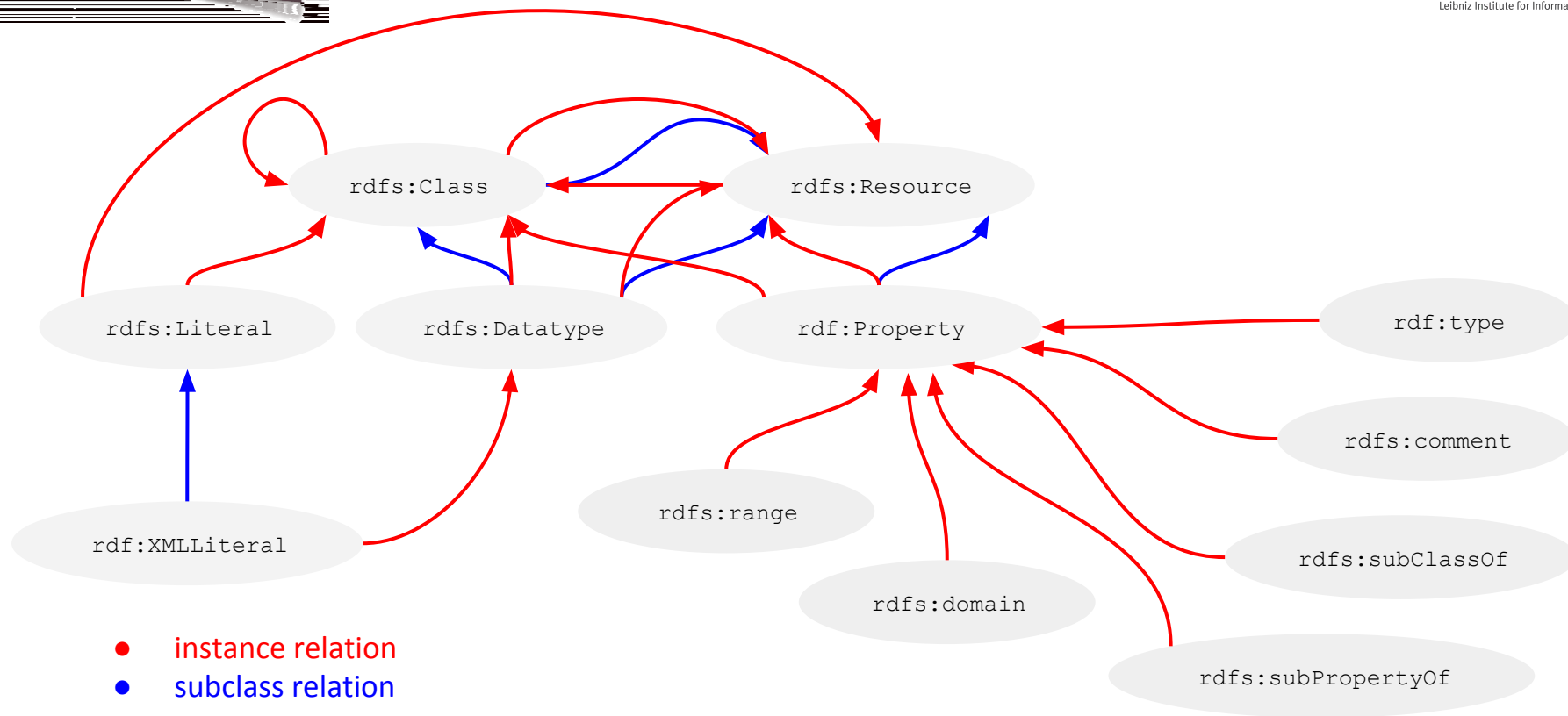


RDF Schema

- Definition of **hierarchical relationships**:
 - **Subclasses** and **superclasses** via `rdfs:subClassOf`
 - Example:
`:Greenhouse_gas rdfs:subClassOf :Air_pollutant .`
 - **Subproperties** and **superproperties** via `subPropertyOf`
 - Example:
`:sublimationTemperature rdfs:subPropertyOf
:temperature .`



(Simplified) RDF Schema Language Model





Some more RDF Schema

- Further RDFS properties:
 - **rdfs:seeAlso**
defines a relation of a resource to another, which explains it
 - **rdfs:isDefinedBy**
subproperty of `rdfs:seeAlso`, defines the relation of a resource to its definition
 - **rdfs:comment**
comment, usually as text
 - **rdfs:label**
„readable“ name of a resource (contrary to ID)



RDF Schema Example

```
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .  
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .  
@prefix owl: <http://www.w3.org/2002/07/owl#>  
@prefix : <http://example.org/Climate#> .
```

```
:Greenhouse_gas      rdf:type      rdfs:Class ;  
                      rdfs:subClassOf :Air_pollutant .  
:Person              rdf:type      rdfs:Class .  
:Scientist           rdfs:subClassOf :Person .  
:Physicist           rdfs:subClassOf :Scientist .  
:Chemist             rdfs:subClassOf :Scientist .
```

Class Definitions

```
:discoverer          rdf:type      rdf:Property ;  
                      rdfs:domain owl:Thing .  
                      rdfs:range   :Person .
```

Property Definitions

```
:Carbon_dioxide      rdf:type      :Greenhouse_gas ;  
                      :discoverer   :Jan_Baptist_van_Helmont ;  
                      :discoverer   :Joseph_Black .  
:Jan_Baptist_van_Helmont rdf:type   :Physicist .  
:Joseph_Black         rdf:type      :Chemist ;  
                      rdfs:label    "Joseph Black"@en ;  
                      rdfs:comment   "co-discovered CO2" .
```

Instance Definitions



Logical Inference with RDF(S)

Next Lecture...

Picture References:

- [1] The Green House Effect, A loose necktie [CC BY-SA]
<https://commons.wikimedia.org/wiki/File:Greenhouse-effect-t445.svg>
- [2] Fourier, in his prefect's garb, Pierre-Claude Gautherot, (1806), [Public Domain]
https://commons.wikimedia.org/wiki/File:Fourier_in_his_coat_of_prefect.jpg?uselang=de
- [3] Shown are the annual global total carbon dioxide emissions by world region since 1750, Hannah Ritchie, Max Roser [CC-BY 4.0]
https://commons.wikimedia.org/wiki/File:Global_annual_CO2_emissions_by_world_region_since_1750.svg
- [4] Benjamin Nowack, *The Semantic Web - Not a Piece of cake...*, at bnode.org, 2009-07-08 , [CC BY 3.0]
<http://bnode.org/blog/2009/07/08/the-semantic-web-not-a-piece-of-cake>
- [5] Trichon, Le grand ballon captif de la cour des Tuileries, 1879, Brown University Library, [Public Domain]
https://commons.wikimedia.org/wiki/File:Le_grand_ballon_captif_de_la_cour_des_Tuileries.jpg